THE UNIVERSITY OF BRITISH COLUMBIA
Faculty of Forestry & Environmental Stewardship

Dear Dr. whatever

Please consider our paper, "Warmer and longer growing seasons increase growth for juvenile trees in a common garden" for publication as a full paper in *New Phytologist*.

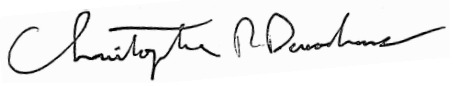
Anthropogenic climate change affects the timing of many biological processes, producing cascading effects. For temperate trees, warming temperatures lengthen their growing season, which should increase growth. However, recent evidence showed that trees may not grow more with warmer seasons which could be detrimental for the critical role that temperate forest carbon sink has on regulating greenhouse gas emissions. Our 'Full paper' adds a rare combination of full growing season length via vegetative phenology and growth via tree rings to address. The favourable context of our common garden pr this fundamental question evidence that under drought- and competition-limited conditions, juvenile trees grew more with longer and warmer seasons. This adds important information to the important question that will determine whether forests regenerate and store carbon with future climate.

What hypotheses or questions does this work address? Why trees do not grow more with longer and warmer seasons? This outstanding question, that we address here in a common garden using juvenile trees of four species sourced from four provenances. We test effects of four different growing season metrics on growth and determined whether provenance latitude changed growth.

How does this work advance our current understanding of plant science? Most recent evidence that growth may not increase with longer seasons was done in natural setting with many confounding factors, which made hard understanding the effect that growing season length had on growth. By a using multi-year, multi-species high quality phenology and growth dataset, we are able to tease apart the contribution of each season metric. We show in particular that the end of season events—most often overlooked—appeared to be more important for growth than spring events.

Why is this work important and timely? Our work provides at a critical time in climate change ecology challenging recent findings on tree growth. We show that young deciduous trees may actually thrive under a warmer climate with has critical implication for forest regeneration and range expansion demonstrating that growing season length and growth may not relate much. Further, we show that the end of season may be more important than the start of season.

All authors contributed to this work and approved this version for submission. The manuscript is 4083 words with a 200 word summary, and 4 figures and is not under consideration elsewhere. We hope you find it suitable for publication in *New Phytologist*, and look forward to hearing from you. Sincerely,



Christophe Rouleau-Desrochers
MSc in Forestry
Department of Forest & Conservation Sciences

The University of British Columbia